Conductance and Cheeger's Inequality

Relaxation · ℓ_1 and ℓ_2^2 metrics · rounding by embedding

CMSC 35410 · Lecture 6

§0 Where we are

Recap: Lectures 2–5

$G = (V, E, w)$ undirected, connected; $d_i = \sum_j w_{ij}$, $\mathrm{vol}(S) = \sum_{i \in S} d_i$, $\pi = d/\mathrm{vol}(V)$. Random-walk Laplacian $L_{rw} = I - D^{-1}A$, spectral gap $\lambda := \lambda_2(L_{rw})$.

Poincaré. $\lambda \,\mathrm{Var}_\pi(f) \le \mathcal{E}(f, f)$, i.e. $\lambda = \min_{f \text{ nonconstant}} \dfrac{\sum_{ij \in E} w_{ij}(f_i - f_j)^2}{\sum_i d_i (f_i - \bar{f})^2}$.

Two certificates for the gap (Lecture 5). *Upper bound on λ:* exhibit a **test function** f. *Lower bound on λ:* route flow along **canonical paths**, $\lambda_2(L) \ge 2n/\kappa$.

Open question from Lecture 5. The binary tree has a *bottleneck at the root* and small λ. Is a small gap *always* caused by a bottleneck? When do test functions and paths meet?

Today

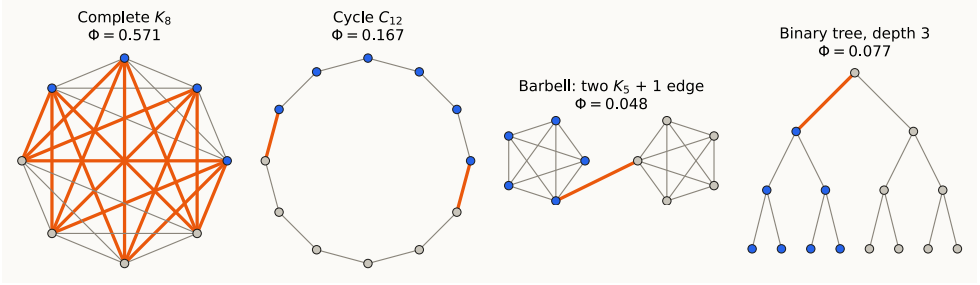
Plan. Conductance is a combinatorial optimization problem. Its **relaxation** is λ. Cheeger's inequality says the relaxation can be **rounded**, and the rounding is a **metric embedding** ($\ell_2^2 \to \ell_1$) followed by a sweep.

1. **Conductance** and a reformulation as an ℓ_1 problem over functions.
2. λ **as the** ℓ_2^2 **relaxation.** The easy direction. Statement of Cheeger. How tight is it?
3. **Geometry:** cut metrics, ℓ_1, and rounding in ℓ_1.
4. **Proof of Cheeger** in three lemmas. Demos: sweep cut, and what it costs to compute the eigenvector.
5. **Beyond:** Leighton–Rao and the landscape of relaxations. (Optional: Cheeger's original manifold version.)

§1 · Conductance

Bottlenecks, quantified

Which of these graphs feel well connected? We want **one number** that says so.



Blue = the set S achieving the minimum, *orange* = the edges it cuts. Small Φ $\Leftrightarrow$ a sparse cut separating two large pieces.

Conductance

For $S \subset V$, the **conductance** of S and of the graph:

$$\Phi(S) = \frac{w(S,\bar{S})}{\min\{\mathrm{vol}(S), \mathrm{vol}(\bar{S})\}}, \qquad \Phi(G) = \min_{\emptyset \neq S \subsetneq V} \Phi(S), \qquad w(S,\bar{S}) = \sum_{i \in S,\, j \notin S} w_{ij}.$$

Random-walk reading. Start the stationary walk, conditioned on being in S (so $i \in S$ w.p. $\propto d_i$). Then $\Pr[\text{leave } S \text{ in one step}] = \dfrac{w(S,\bar{S})}{\mathrm{vol}(S)}$.

So $\Phi(S)$ is the **escape probability** of the set (for the smaller side), and $\Phi(G)$ is the worst bottleneck. In the notation of Lecture 2:
$Q(S,\bar{S}) / \min\{\pi(S), \pi(\bar{S})\}$ with $Q = w/\mathrm{vol}(V)$.

$\Phi(G)$ small $\Rightarrow$ a set the walk rarely leaves $\Rightarrow$ slow mixing. **Is the converse true?**

Examples

Graph (n vertices)	$\Phi(G)$	$\lambda = \lambda_2(L_{rw})$
Complete K_n	$\lceil n/2 \rceil/(n-1) \approx \frac{1}{2}$	$\dfrac{n}{n-1}$
Cycle C_n (n even)	$\dfrac{2}{n}$	$1 - \cos\dfrac{2\pi}{n} \approx \dfrac{2\pi^2}{n^2}$
Hypercube Q_d ($n = 2^d$)	$\dfrac{1}{d} = \dfrac{1}{\log_2 n}$	$\dfrac{2}{d}$
Binary tree ($n = 2^{h+1} - 1$)	$\dfrac{1}{n-2}$ (cut one root edge)	$\Theta(1/n)$ (numerically $\approx 0.5/n$)
Barbell (two K_m + 1 edge)	$\dfrac{1}{m(m-1)+1}$	$\approx \Phi$

Look at the cycle. $\Phi \approx 2/n$ but $\lambda \approx 2\pi^2/n^2$. The gap is about the **square** of the conductance. Hold on to this: it is the square root in Cheeger's inequality.

Conductance as an ℓ_1 problem over functions

Define, for $f \in \mathbb{R}^V$: $\|\nabla f\|_1 = \sum_{ij\in E} w_{ij}|f_i - f_j|$ and the ℓ_1 **variance** $\mathrm{Var}_1(f) = \min_c \sum_i d_i|f_i - c|$ (deviation from the π-median).

Claim. $\Phi(G) = \min_{f \text{ nonconstant}} \dfrac{\|\nabla f\|_1}{\mathrm{Var}_1(f)}$, and the minimum is attained by a cut vector $f = \mathbf{1}_S$.

($\leq$) For $f = \mathbf{1}_S$: $\|\nabla f\|_1 = w(S, \bar{S})$ and $\mathrm{Var}_1(f) = \min_{c\in[0,1]} \big(\mathrm{vol}(S)(1-c) + \mathrm{vol}(\bar{S})c\big) = \min\{\mathrm{vol}(S), \mathrm{vol}(\bar{S})\}$.

($\geq$) Let $S_t = \{f > t\}$. Layer-cake: $|f_i - f_j| = \int_{\mathbb{R}} \big|\mathbf{1}_{S_t}(i) - \mathbf{1}_{S_t}(j)\big|\, dt$, so $\|\nabla f\|_1 = \int w(S_t, \bar{S}_t)\, dt$. With c the median, $\mathrm{Var}_1(f) = \int \min\{\mathrm{vol}(S_t), \mathrm{vol}(\bar{S}_t)\}\, dt$. A ratio of integrals is at least the smallest ratio of integrands:

$$\|\nabla f\|_1 = \int \Phi(S_t)\, \min\{\mathrm{vol}(S_t), \mathrm{vol}(\bar{S}_t)\}\, dt \;\geq\; \min_t \Phi(S_t) \cdot \mathrm{Var}_1(f). \;\square$$

Symmetrized conductance $\bar{\phi}$, and swapping ℓ_1 for ℓ_2^2

(a) Fix the denominator. The median is awkward. Use the **pairwise** spread instead, the ℓ_1 analogue of $\mathrm{Var} = \frac{1}{2}\mathbb{E}(X-Y)^2$:

$$\mathrm{Spr}_1(f) = \frac{1}{\mathrm{vol}(V)} \sum_{i<j} d_i d_j |f_i - f_j|.$$

On $\mathbf{1}_S$ it equals $\mathrm{vol}(S)\mathrm{vol}(\bar{S})/\mathrm{vol}(V) \in \left[\frac{1}{2}, 1\right] \cdot \min\{\mathrm{vol}(S), \mathrm{vol}(\bar{S})\}$. The same layer-cake argument gives

$$\bar{\phi}(G) := \min_f \frac{\|\nabla f\|_1}{\mathrm{Spr}_1(f)} = \min_S \bar{\phi}(S), \quad \bar{\phi}(S) = \frac{w(S,\bar{S})\,\mathrm{vol}(V)}{\mathrm{vol}(S)\mathrm{vol}(\bar{S})}, \qquad \Phi \le \bar{\phi} \le 2\Phi.$$

(b) Swap the norm. Replace $|f_i - f_j|$ by $(f_i - f_j)^2$ in numerator **and** denominator. (Numerator too: the ratio must stay scale-invariant. The two agree on $\{0,1\}$-vectors.)

	ℓ_1	ℓ_2^2
numerator	$\sum_E w\,\lvert\Delta f\rvert$	$\sum_E w\,(\Delta f)^2$
centered / pairwise	Var_1 (median) / Spr_1	Var_2 (mean) $= \mathrm{Spr}_2$
denominator	(differ by ≤ 2)	**(identical)**
minimum over f	Φ / $\bar{\phi}$, attained at cuts	λ, attained at an eigenvector
difficulty	NP-hard	linear algebra

Computational status

- Minimizing conductance (sparsest cut) is **NP-hard**.
- Best known polynomial-time approximation: $O(\sqrt{\log n})$ (Arora–Rao–Vazirani, 2004). Earlier: $O(\log n)$ (Leighton–Rao, 1988).
- Under the Unique Games Conjecture, **no constant factor** is possible (Chawla–Krauthgamer–Kumar–Rabani–Sivakumar).

So relax. Find a *fractional* solution that is easy to compute and gives a *lower bound*, then **round** it to a real cut and bound the loss.

§2 · λ as a relaxation

The relaxation: from cut vectors to all of $\mathbb{R}^V$

On a cut vector the ℓ_1 and ℓ_2^2 quantities coincide, since $|a| = a^2$ for $a \in \{0, \pm 1\}$:

$$\mathbf{1}_S^\top L\, \mathbf{1}_S = w(S, \bar{S}), \qquad \mathrm{Spr}_2(\mathbf{1}_S) = \frac{\mathrm{vol}(S)\mathrm{vol}(\bar{S})}{\mathrm{vol}(V)}.$$

Hence

$$\bar{\phi}(G) = \min_{x \in \{0,1\}^V \setminus \{\mathbf{0}, \mathbf{1}\}} \frac{\sum_E w_{ij}(x_i - x_j)^2}{\mathrm{Var}_2(x)} \quad \longrightarrow \quad \min_{x \in \mathbb{R}^V \text{ nonconstant}} \frac{\sum_E w_{ij}(x_i - x_j)}{\mathrm{Var}_2(x)}$$

Drop integrality and you get the spectral gap. Same objective, larger feasible set. The new problem is an eigenvalue problem (Poincaré, Lecture 2), solved exactly by the Fiedler vector.

The easy direction

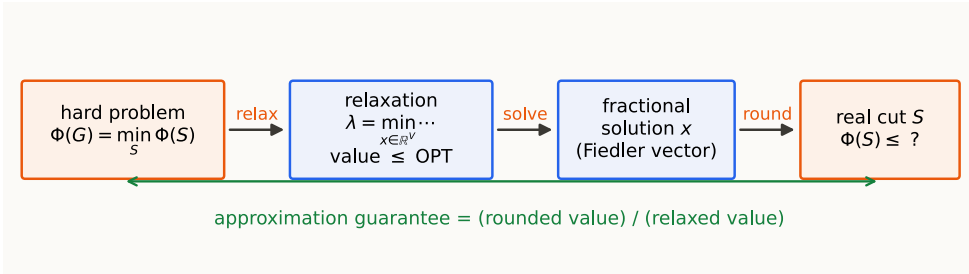
Theorem (easy side). $\lambda \le \bar{\phi}(G) \le 2\,\Phi(G)$, i.e. $\frac{1}{2}\lambda \le \Phi(G)$.

Proof. Cut vectors are feasible for the relaxation, so the relaxed minimum is smaller: $\lambda \le \bar{\phi}(S)$ for every S. Then $\bar{\phi} \le 2\Phi$. $\square$

Reading. If G has a sparse cut S, then $x = \mathbf{1}_S$ is a **test function** with small Rayleigh quotient, so λ is small and the walk mixes slowly: *bottlenecks cause small gaps*. (This is the Lecture 5 test-function argument, with a cut as the test function.)

Contrapositive: **if λ is large, every cut is large.** The gap *certifies* expansion, in polynomial time.

Template: relaxation, then rounding



The same pattern runs through this course: **Leighton–Rao, ARV, k-way Cheeger, SDPs**. What changes is the relaxation and the rounding.

Cheeger's inequality

Theorem (Cheeger; Alon–Milman, Dodziuk). For every connected weighted graph,

$$\tfrac{1}{2}\lambda \;\le\; \Phi(G) \;\le\; \sqrt{2\lambda}.$$

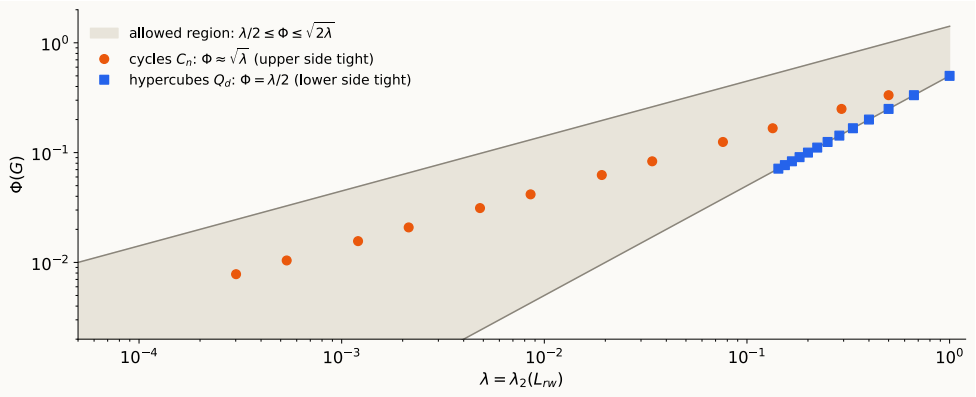
Moreover, the sweep cuts of the Fiedler vector contain a set S with $\Phi(S) \le \sqrt{2\lambda}$.

- Left: the easy side (relaxation). Done.
- Right: the **rounding**. Given a vector $x \in \mathbb{R}^V$, produce a set S.

Equivalently: $\lambda \;\ge\; \Phi^2/2$. In the notation of the continuous version: $\lambda \ge h^2/4$.

Before the proof: **is this sandwich tight? Can the square root be avoided?**

Both sides of Cheeger are tight



§3 · Geometry: metrics, cut metrics, ℓ_1

Metrics and "volume"

A **semimetric** is a symmetric $d : V \times V \to \mathbb{R}_{\geq 0}$ with $d_{ii} = 0$. A **metric** also satisfies the triangle inequality $d_{ij} \leq d_{ik} + d_{kj}$. For a semimetric d define

$$d(G) = \sum_{ij \in E} w_{ij} d_{ij}, \qquad d(K_G) = \sum_{i<j} \frac{d_i d_j}{\mathrm{vol}(V)} d_{ij},$$

the "total length" of d on G and on the complete graph K_G with weights $d_i d_j / \mathrm{vol}(V)$.

Cut metric of S: $\delta^S_{ij} = \mathbf{1}[\, i, j \text{ on different sides of } S \,]$. Then

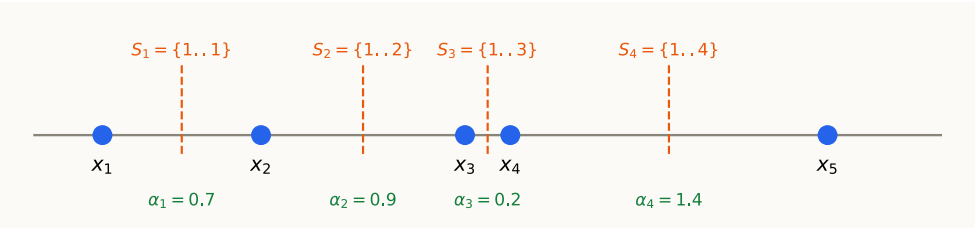
$$\delta^S(G) = w(S, \bar{S}), \qquad \delta^S(K_G) = \frac{\mathrm{vol}(S)\mathrm{vol}(\bar{S})}{\mathrm{vol}(V)} \qquad \Longrightarrow \qquad \bar{\phi}(S) = \frac{\delta^S(G)}{\delta^S(K_G)}.$$

Metric form of the objective. $\bar{\phi}(G) = \min_S \dfrac{\delta^S(G)}{\delta^S(K_G)}$. Moreover, a function f gives the *line metric* $d_{ij} = |f_i - f_j|$ with $d(G) = \|\nabla f\|_1$ and $d(K_G) = \mathrm{Spr}_1(f)$: slide 7 was already a statement about metrics.

The cut cone is ℓ_1

A metric d is ℓ_1**-embeddable** if $d_{ij} = \|v_i - v_j\|_1$ for some $v_i \in \mathbb{R}^k$. The **cut cone** is $\text{CUT}_V = \{\sum_S \alpha_S \, \delta^S : \ \alpha_S \geq 0\}$.

Theorem. A semimetric on V is ℓ_1-embeddable $\iff$ it lies in the cut cone.



$$|x_i - x_j| = \sum_t \alpha_t \, \delta_{ij}^{S_t}, \qquad S_t = \{1, \ldots, t\}, \ \alpha_t = x_{t+1} - x_t \quad \text{(points sorted: } x_1 \leq \ldots \leq$$

Proof: cut cone $= \ell_1$

(⇐) A cut metric is ℓ_1-embeddable: $\delta^S_{ij} = |\mathbf{1}_S(i) - \mathbf{1}_S(j)|$. A nonnegative combination is embedded by *concatenating* the coordinates, scaling the S-th one by α_S: $\|\cdot\|_1$ adds over coordinates.

(⇒) Suppose $d_{ij} = \sum_k |v_i^{(k)} - v_j^{(k)}|$. Both sides are additive over coordinates, so it suffices to treat one coordinate $x = v^{(k)} \in \mathbb{R}^V$. Order the vertices so that $x_1 \le \dots \le x_n$ and put $S_t = \{1, \dots, t\}$. For $i < j$,

$$|x_i - x_j| = x_j - x_i = \sum_{t=i}^{j-1} (x_{t+1} - x_t) = \sum_{t=1}^{n-1} (x_{t+1} - x_t)\, \delta^{S_t}_{ij},$$

because $\delta^{S_t}_{ij} = 1$ exactly when $i \le t < j$. So $|x_i - x_j|$ is a nonnegative combination of cut metrics (the same layer-cake as slide 6). Summing over coordinates, $d \in \mathrm{CUT}_V$. □

Why ℓ_1 is the right geometry for rounding. A point set on a line *is* a family of nested cuts $S_1 \subset S_2 \subset \cdots$. To round an ℓ_1 solution, look at its cuts.

Lemma 1: rounding an ℓ_1 solution

For $x \in \mathbb{R}^V_{\geq 0}$ let $R_1(x) = \dfrac{\sum_{ij \in E} w_{ij}|x_i - x_j|}{\sum_i d_i |x_i|}$ and $S_t = \{i : x_i > t\}$.

Lemma 1 (sweep). There is a $t \geq 0$ with $S_t \neq \emptyset$ and $\dfrac{w(S_t, \bar{S}_t)}{\mathrm{vol}(S_t)} \leq R_1(x)$.

Proof. Layer-cake identities, valid because $x \geq 0$:

$$\sum_{ij \in E} w_{ij}|x_i - x_j| = \int_0^\infty w(S_t, \bar{S}_t)\, dt, \qquad \sum_i d_i x_i = \int_0^\infty \mathrm{vol}(S_t)\, dt.$$

Let t^* minimize $w(S_t, \bar{S}_t)/\mathrm{vol}(S_t)$ over t with $S_t \neq \emptyset$. Then

$$\int_0^\infty w(S_t, \bar{S}_t)\, dt = \int_0^\infty \frac{w(S_t, \bar{S}_t)}{\mathrm{vol}(S_t)} \mathrm{vol}(S_t)\, dt \;\geq\; \frac{w(S_{t^*}, \bar{S}_{t^*})}{\mathrm{vol}(S_{t^*})} \int_0^\infty \mathrm{vol}(S_t)\, dt. \;\square$$

Remark. The same "ratio of sums $\geq$ smallest ratio" argument shows that *any* ℓ_1 metric d has a cut with $\bar{\phi}(S) \leq d(G)/d(K_G)$. **Rounding an ℓ_1 solution is free**: no loss at all.

λ as an ℓ_2^2 relaxation

A semimetric g is **ℓ_2^2-embeddable** if $g_{ij} = \|v_i - v_j\|_2^2$ for some $v_i \in \mathbb{R}^k$.

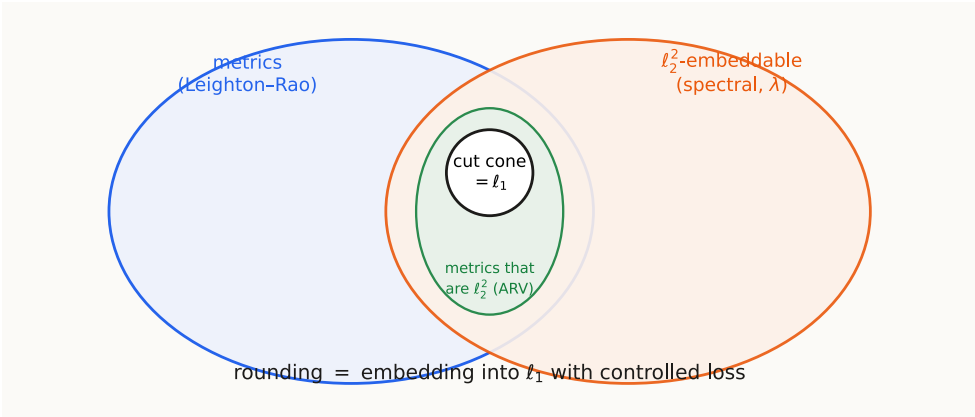
Lemma. $\lambda = \min\left\{ \dfrac{g(G)}{g(K_G)} : \ g\ \ell_2^2\text{-embeddable},\ g \neq 0 \right\}$. Cut metrics are ℓ_2^2-embeddable ($\delta_{ij}^S = (\mathbf{1}_S(i) - \mathbf{1}_S(j))^2$), so this is a relaxation of $\bar{\phi}$.

Proof. ($\leq$) One-dimensional embeddings $v_i = x_i \in \mathbb{R}$ give $g(G)/g(K_G) = \sum_E w(\Delta x)^2 / \mathrm{Var}_2(x)$, minimized at λ. ($\geq$) For $v_i \in \mathbb{R}^k$, split into coordinates and use the **mediant inequality** $\dfrac{\sum_u a_u}{\sum_u b_u} \geq \min_u \dfrac{a_u}{b_u}$:

$$\frac{g(G)}{g(K_G)} = \frac{\sum_{u=1}^k \sum_E w_{ij}(v_i^u - v_j^u)^2}{\sum_{u=1}^k \mathrm{Var}_2(v^u)} \ \geq\ \min_u \frac{\sum_E w_{ij}(v_i^u - v_j^u)^2}{\mathrm{Var}_2(v^u)} \ \geq\ \lambda. \ \square$$

Remarks. (i) Higher-dimensional embeddings add nothing: the relaxation is the same as for scalars. (ii) An ℓ_2^2 semimetric need *not* satisfy the triangle inequality (three collinear points: $1^2 + 1^2 < 2^2$ fails). It is not a metric.

Nested relaxations of the cut cone



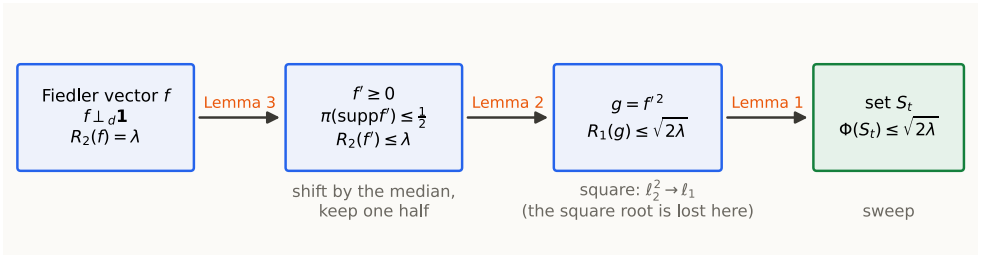
Both relaxations **enlarge** the cut cone: optimizing over a bigger set gives a **smaller** value (a lower bound). The price is paid at rounding, when we must come back down to ℓ_1.

§4 · Proof of Cheeger’s inequality

Plan: three lemmas

Let $R_2(x) = \dfrac{\sum_E w_{ij}(x_i - x_j)^2}{\sum_i d_i x_i^2}$ and $R_1(x) = \dfrac{\sum_E w_{ij}|x_i - x_j|}{\sum_i d_i |x_i|}$ (uncentered).

Let f be the Fiedler vector: $f \perp_d \mathbf{1}$ and $R_2(f) = \lambda$.



Lemma 1 (slide 16) is already proved. Now Lemmas 2 and 3.

Lemma 2: $\ell_2^2 \to \ell_1$ by squaring

Lemma 2. For $x \geq 0$: $R_1(x^2) \leq \sqrt{2R_2(x)}$, where $(x^2)_i = x_i^2$.

Proof. Since $x \geq 0$, $|x_i^2 - x_j^2| = |x_i - x_j|(x_i + x_j)$. By Cauchy–Schwarz and $(a+b)^2 \leq 2(a^2+b^2)$:

$$\sum_E w_{ij}|x_i^2 - x_j^2| \leq \sqrt{\sum_E w_{ij}(x_i - x_j)^2}\,\sqrt{\sum_E w_{ij}(x_i + x_j)^2} \leq \sqrt{R_2(x)\textstyle\sum_i d_i x_i^2}\,\sqrt{2\sum_i}$$

(We used $\sum_E w_{ij}(x_i^2 + x_j^2) = \sum_i d_i x_i^2$.) Divide by $\sum_i d_i x_i^2 = \sum_i d_i |x_i^2|$. $\square$

This is where the square root is lost. The map $x \mapsto x^2$ turns an ℓ_2^2 object into an ℓ_1 one, but it is **not a distortion-bounded embedding of all pairs**. It preserves the *ratio* of the two sums only up to a square root, via Cauchy–Schwarz. On the cycle (slide 13) the eigenvector is a cosine and Cauchy–Schwarz is nearly tight: that is the example showing the loss is real.

Lemma 3: a nonnegative vector with half the support

Lemma 3. If $f \perp_d \mathbf{1}$, there is $f' \geq 0$ with $\pi(\{f' > 0\}) \leq \frac{1}{2}$ and $R_2(f') \leq R_2(f)$.

Proof. Let m be a π-median of f and $\bar{f} = f - m\mathbf{1}$, so $\pi(\bar{f} > 0) \leq \frac{1}{2}$ and $\pi(\bar{f} < 0) \leq \frac{1}{2}$. The numerator is unchanged. Since $\sum_i d_i f_i = 0$, $\sum_i d_i \bar{f}_i^2 = \sum_i d_i f_i^2 + m^2 \mathrm{vol}(V) \geq \sum_i d_i f_i^2$, so $R_2(\bar{f}) \leq R_2(f)$.

Write $\bar{f} = a - b$ with $a = \max\{\bar{f}, 0\}$, $b = \max\{-\bar{f}, 0\}$: **disjointly supported** and ≥ 0. Then for each edge ij, with $x = a_i - a_j$, $y = b_i - b_j$: $xy = -(a_i b_j + a_j b_i) \leq 0$, so

$$(x-y)^2 \;\geq\; x^2+y^2 \implies N(a-b) \geq N(a)+N(b), \qquad D(a-b) = D(a)+D(b).$$

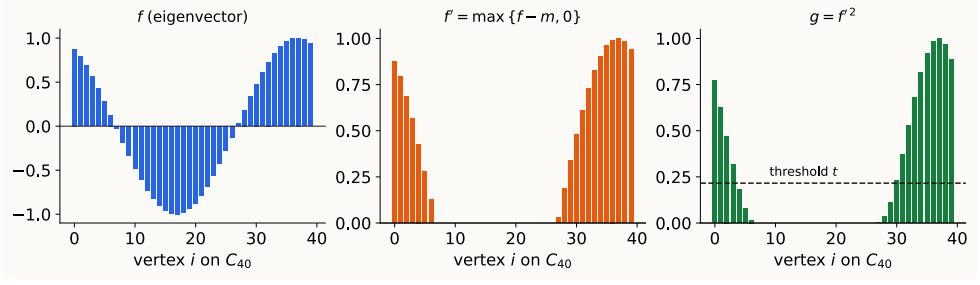
Hence $R_2(\bar{f}) \geq \dfrac{N(a)+N(b)}{D(a)+D(b)} \geq \min\{R_2(a), R_2(b)\}$. Take f' to be the better of a, b. $\square$

Putting it together

Proof of Cheeger. Fiedler $f \xrightarrow{\text{L3}} f' \xrightarrow{\text{L2}} g = f'^2 \xrightarrow{\text{L1}} S_t$:

$$\Phi(S_t) = \frac{w(S_t, \bar{S}_t)}{\mathrm{vol}(S_t)} \;\le\; R_1(g) \;\le\; \sqrt{2R_2(f')} \;\le\; \sqrt{2\lambda}.$$

The first equality holds because $S_t \subseteq \operatorname{supp} f'$ has $\pi(S_t) \le \frac{1}{2}$, so $\mathrm{vol}(S_t) = \min\{\mathrm{vol}(S_t), \mathrm{vol}(\bar{S}_t)\}$. $\square$



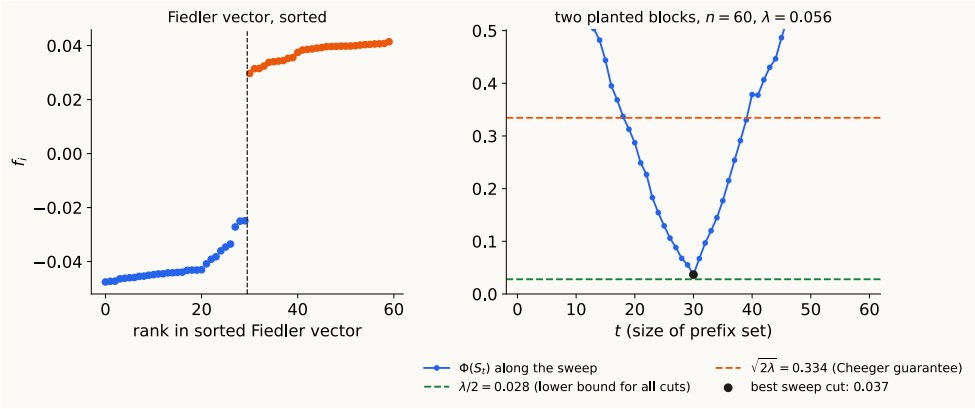
Algorithm: spectral partitioning

1. Compute (approximately) the Fiedler vector f of L_{rw}.
2. Sort the vertices by f; evaluate $\Phi(S_t)$ for the $n-1$ prefix sets.
3. Return the best prefix (or the best of the two sides' prefixes).

Guarantee, with a two-sided certificate. Output S satisfies $\Phi(S) \le \sqrt{2\lambda} \le 2\sqrt{\Phi(G)}$, and *every* cut has $\Phi \ge \lambda/2$. So the algorithm both finds a cut and **proves** how far from optimal it can be: ratio $\le \sqrt{8/\lambda}$ relative to the lower bound $\lambda/2$.

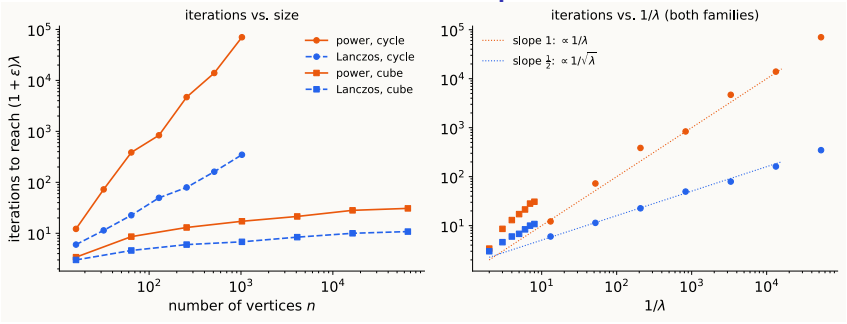
Each iteration of power method or Lanczos costs $O(m)$. **How many iterations?** Next two slides.

Demo: the sweep cut on a graph with a bottleneck



The best sweep cut lies **between** $\lambda/2$ and $\sqrt{2\lambda}$, as the theorem promises (here it recovers the planted bisection).

Demo: how fast can we compute the Fiedler vector?



$\varepsilon = 0.1$, average of 5 random starts. We measure the **Rayleigh quotient**, not eigenvector error: λ is repeated on both graphs (multiplicity 2 on the cycle, d on the cube), and Cheeger only needs a vector with a small quotient.

Takeaway. Standard methods have running time depending on λ: power $\approx 1/\lambda$, Krylov (Lanczos; conjugate gradient is its linear-system sibling) $\approx 1/\sqrt{\lambda}$. Cycle: n^2 vs. n. Cube: $\log n$ vs. $\sqrt{\log n}$. **The graphs with the sparsest cuts are exactly the slowest to cut.** Acceleration helps but does not remove the dependence. (Next: electrical flows and solvers.)

§5 · Beyond: other relaxations

Leighton–Rao: relax to all metrics

Keep the triangle inequality, drop ℓ_1-ness:

$$\min\ d(G) \quad \text{s.t.} \quad d(K_G) = 1, \quad d \text{ a metric on } V.$$

A **linear program** (the triangle inequalities are the constraints). It is a relaxation because $\text{CUT}_V \subseteq \{\text{metrics}\}$.

Dual = *maximum concurrent multicommodity flow*: route $\alpha\, d_i d_j / \text{vol}(V)$ units between every pair i, j with edge congestion $\le w_{ij}$, maximize α. This is exactly the **canonical-path** congestion from Lecture 5 (there with one fixed routing, here optimized).

Rounding by embedding. Bourgain (1985): every n-point metric embeds into ℓ_1 with distortion $O(\log n)$. Then use Lemma 1 / the cut cone: $O(\log n)$**-approximation** (Linial–London–Rabinovich, Aumann–Rabani).

Second instance of the template: **relax** $\to$ **solve** $\to$ **embed into** ℓ_1 $\to$ **sweep.**

The landscape

Relaxation	Feasible set	Solved by	Rounding	Guarantee
Spectral	ℓ_2^2-embeddable (no triangle ineq.)	eigenvector	square, sweep (L3, L2, L1)	$\Phi(S) \le \sqrt{2\lambda} \le 2\sqrt{\Phi^*}$
Leighton–Rao	all metrics	LP	Bourgain $\to \ell_1$, sweep	$O(\log n)$
ARV	ℓ_2^2 **and** triangle inequality	SDP	structure theorem + sweep	$O(\sqrt{\log n})$

- Spectral is **dimension-free** (no $\log n$) but loses a **square root in the value**: great when Φ^* is a constant, poor when Φ^* is tiny.
- LR and ARV lose only in the *dimension*, at the price of an LP/SDP.
- ARV sits in the intersection of the two sets on slide 18: next weeks' SDP lecture.

Preview: higher-order Cheeger

What if the graph has k **clusters** rather than one bottleneck? Then $\lambda_2, \dots, \lambda_k$ are all small, and one eigenvector is not enough.

- **Relaxation:** k orthogonal test functions, i.e. a map $V \to \mathbb{R}^k$ ("spectral embedding").
- **Rounding:** partition the *embedded point cloud* in $\mathbb{R}^k$: an embedding into ℓ_1 or ℓ_2 with controlled distortion, plus a clustering step.
- **Result** (Lee–Oveis Gharan–Trevisan; Louis–Raghavendra–Tetali–Vempala): $\dfrac{\lambda_k}{2} \le \rho_k(G) \le O(\mathrm{poly}(k))\sqrt{\lambda_k}$, where ρ_k is the best k-way conductance.

Same template, now with k-dimensional embeddings. **Next lecture.**

Takeaways

- $\Phi(G)$ is NP-hard. λ **is its relaxation**: replace ℓ_1 by ℓ_2^2 and $\{0,1\}^V$ by $\mathbb{R}^V$.
- **Easy side:** cut vectors are feasible, so $\lambda \leq \bar{\phi} \leq 2\Phi$.
- **Hard side:** shift by the median (L3), **square** to pass from ℓ_2^2 to ℓ_1 (L2), **sweep** (L1). Loss: a square root.
- The $\sqrt{}$ is **tight**: cycle vs. hypercube.
- Standard eigenvector methods run in time depending on λ (power $1/\lambda$, Krylov $1/\sqrt{\lambda}$): bottlenecks are exactly what makes them slow.
- Other relaxations (LR, ARV) differ only in the feasible set and the embedding used to round.

Reading. Trevisan's posts on Cheeger in manifolds and its proof; Erasmo Tani's note on the continuous Cheeger inequality.

Optional: Cheeger’s original (manifolds)

Cheeger's setting (1970)

$\mathcal{M}$ compact Riemannian manifold without boundary, measure μ, Laplacian $L = -\operatorname{div}\nabla$ (the same operator whose heat flow we studied). Cheeger constant:

$$h(\mathcal{M}) = \inf_{0<\mu(S)\le \frac{1}{2}\mu(\mathcal{M})} \frac{\mu_{n-1}(\partial S)}{\mu(S)}.$$

Graph	Manifold
vertices V, weights d_i	points $\mathcal{M}$, measure μ
$\operatorname{vol}(S)$	$\mu(S)$
cut $w(S, \bar{S})$	boundary area $\mu_{n-1}(\partial S)$
$L = \nabla^*\nabla$	$L = -\operatorname{div}\nabla$
$\sum_E w(\Delta f)^2 = f^\top L f$	$\int \|\nabla f\|^2 d\mu$
$\Phi(G)$	$h(\mathcal{M})$

Theorem (Cheeger). $\lambda_1 \ge h^2/4$, with λ_1 the first nonzero eigenvalue. (This is the graph inequality $\Phi \le \sqrt{2\lambda}$.)

Same three lemmas, calculus replaces sums

With $R_1(f) = \dfrac{\int \|\nabla f\| d\mu}{\int |f| d\mu}$ and $R_2(f) = \dfrac{\int \|\nabla f\|^2 d\mu}{\int f^2 d\mu}$:

- **Lemma 1.** The **coarea formula** $\int \|\nabla f\| d\mu = \int_0^\infty \mu_{n-1}(\{f = t\}) dt$ replaces the layer-cake identity: some superlevel set has $\mu_{n-1}(\partial S_t)/\mu(S_t) \le R_1(f)$.
- **Lemma 2.** The **chain rule** $\nabla f^2 = 2f\nabla f$ and Cauchy–Schwarz give $R_1(f^2) \le 2\sqrt{R_2(f)}$. The constant is 2 rather than $\sqrt{2}$: nothing is lost in an edge-wise $(a+b)^2 \le 2(a^2+b^2)$.
- **Lemma 3.** Shift by the median; the gradients of disjointly supported functions are orthogonal pointwise, so now $N(a-b) = N(a) + N(b)$ **exactly**.

Result: $h \le 2\sqrt{\lambda_1}$, i.e. $\lambda_1 \ge h^2/4$.

Remark (Buser). The reverse inequality, $\lambda_1 \le C(h + h^2)$, needs a lower bound on Ricci curvature, whereas for graphs the easy side is free. (Discussed in the lecture notes.)

Appendix: definitions for the board

Graph, conductance, relaxation

Graph. $G=(V,E,w)$, $d_i=\sum_j w_{ij}$, $\mathrm{vol}(S)=\sum_{i\in S} d_i$, $\pi=d/\mathrm{vol}(V)$, $w(S,\bar S)=\sum_{i\in S, j\notin S} w_{ij}$.

Conductance. $\Phi(S)=\dfrac{w(S,\bar S)}{\min\{\mathrm{vol}S,\mathrm{vol}\bar S\}}$; $\bar\phi(S)=\dfrac{w(S,\bar S)\,\mathrm{vol}(V)}{\mathrm{vol}(S)\mathrm{vol}(\bar S)}$; $\Phi\le\bar\phi\le 2\Phi$.

Variances / spreads. $\mathrm{Var}_1(f)=\min_c\sum_i d_i|f_i-c|$; $\mathrm{Var}_2(f)=\sum_i d_i(f_i-\bar f)^2=\mathrm{Spr}_2(f)$; $\mathrm{Spr}_p(f)=\frac{1}{\mathrm{vol}(V)}\sum_{i<j} d_i d_j|f_i-f_j|^p$.

Gap. $\lambda=\lambda_2(L_{rw})=\min_f\dfrac{\sum_E w(\Delta f)^2}{\mathrm{Var}_2(f)}$. **Cheeger:** $\frac12\lambda\le\Phi(G)\le\sqrt{2\lambda}$.

Metrics and lemmas

Metric view. $d(G) = \sum_E w_{ij} d_{ij}$; $d(K_G) = \sum_{i<j} \frac{d_i d_j}{\mathrm{vol} V} d_{ij}$; $\delta^S(G) = w(S, \bar{S})$, $\delta^S(K_G) = \frac{\mathrm{vol} S \, \mathrm{vol} \bar{S}}{\mathrm{vol} V}$. $\mathrm{CUT}_V = \ell_1$-embeddable metrics.

Rayleigh-type quotients (uncentered). $R_1(x) = \frac{\sum_E w|x_i - x_j|}{\sum_i d_i |x_i|}$, $R_2(x) = \frac{\sum_E w (x_i - x_j)^2}{\sum_i d_i x_i^2}$.

Lemma 1 (sweep). $x \geq 0 \Rightarrow \exists t : \; w(S_t, \bar{S}_t)/\mathrm{vol}(S_t) \leq R_1(x)$. **Lemma 2.** $x \geq 0 \Rightarrow R_1(x^2) \leq \sqrt{2 R_2(x)}$. **Lemma 3.** $f \perp_d \mathbf{1} \Rightarrow \exists f' \geq 0$, $\pi(f' > 0) \leq \frac{1}{2}$, $R_2(f') \leq R_2(f)$.

Mediant inequality. $\dfrac{\sum a_u}{\sum b_u} \geq \min_u \dfrac{a_u}{b_u}$ for $b_u > 0$.

Layer cake. $|x_i - x_j| = \int |\mathbf{1}[x_i > t] - \mathbf{1}[x_j > t]| \, dt$; for $x \geq 0$, $\sum_i d_i x_i = \int_0^\infty \mathrm{vol}(S_t) dt$.